

Learnability Boundary: How Complex Can Neural Networks Learn Pseudo-Random Sequences?

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Abstract: Classical information theory assumes observers have unlimited computational power, making pseudo-random numbers “low entropy” (seeds are only a few hundred bits). However, for computationally bounded neural networks, the same data may be “completely unlearnable noise.” This paper experimentally explores the boundary of neural network learnability: from simple periodic patterns to cryptographically secure random numbers, we identify where Epiplexity (extractable structure) transitions from non-zero to zero. We train identical Transformer architectures on 6 sequences of varying complexity and find: (1) pseudo-random sequences with state space 2^{256} (simple LCG, 5-bit LFSR) can be perfectly learned; (2) pseudo-random with state space 2^{31} (glibc LCG) is completely unlearnable, with test accuracy equal to random guessing; (3) 31-bit LFSR exhibits **partial Grokking**—the model learns 1 bit of the sequence’s pattern (50% accuracy) but cannot learn the remaining 7 bits. These findings validate the core prediction of the Epiplexity paper: **information is observer-dependent; the same data presents different learnability to observers with different computational power.**

1. Introduction: The 78-Year Blind Spot in Information Theory

In 1948, when Claude Shannon founded information theory, he assumed observers have **unlimited computational power**. Under this assumption, the output of a cryptographic random number generator has “low information”—because the program generating it is only a few lines, and the seed is only 256 bits.

But this assumption never holds in reality.

For a computationally bounded neural network, that same 1GB of pseudo-random data may be “completely incomprehensible pure noise”—it cannot find any pattern no matter how much computation it expends.

In January 2026, a paper from CMU and NYU, “From Entropy to Epiplexity,” formally challenged this 78-year-old assumption and proposed a new framework:

$$\text{Data} = \text{Epiplexity}(S_T) + \text{Time-bounded Entropy}(H_T)$$

Where: - **Epiplexity** (S_T): The structure/patterns you can extract from data given computational budget T - **Time-bounded Entropy** (H_T): The part you simply cannot learn

Core insight: For the same data, observers with more computational power can see more structure (higher Epiplexity) and less noise (lower Entropy).

This paper aims to **experimentally validate this theory**: using neural networks with identical architecture to learn pseudo-random sequences of different complexity, finding where the learnability boundary lies.

2. Experimental Design

2.1 Core Question

How “weak” can pseudo-random be for neural networks to still learn it?

We designed 6 sequence generators ranging from simple to complex:

Sequence Type	Generation Rule	State Space	Expected
periodic	$(i \bmod 7) > 3 \rightarrow 0/1$	7	Yes
lcg_simple	$x_{n+1} = (3x_n + 7) \bmod 256$	256	Yes
lfsr_5	$x_n = x_{n-1} \oplus x_{n-5}$	32	Yes
lcg_glibc	$x_{n+1} = (1103515245x_n + 12345) \bmod 2^{31}$	2^{31}	Boundary
lfsr_31	31-bit LFSR (polynomial $x^{31} + x^3 + 1$)	2^{31}	Boundary
csprng	Python <code>secrets.randbits</code> (cryptographically secure)	∞	No

Task: Given the first 16 tokens of a sequence, predict the 17th.

2.2 Model Configuration

We reuse the architecture from our previous Grokking experiments:

Parameter	Value
Model	2-layer Transformer
Hidden dim	128
Attention heads	4
Context length	16
Optimizer	AdamW
Learning rate	1e-3
Weight decay	1.0
Total steps	100,000
Training ratio	30%
Total sequence length	10,000

2.3 Evaluation Metrics

- **Test accuracy:** Evaluated on 70% of the data
- **Random baseline:** periodic/lfsr_5 = 50% (binary classification), others = $1/256 \approx 0.4\%$
- **Learnability criterion:** Test accuracy significantly above random baseline

3. Experimental Results

3.1 Main Result: Phase Transition Boundary Between 256 and 2^{31}

Sequence Type	Final Test Accuracy	Random Baseline	Learnable?
periodic	100.00%	50.00%	Yes
lcg_simple	100.00%	0.39%	Yes
lfsr_5	100.00%	50.00%	Yes
lcg_glibc	0.41%	0.39%	No
lfsr_31	50.16%	0.39%	Partial
csprng	0.33%	0.39%	No

Key Findings:

1. **Sequences with state space 2^{256} can be perfectly learned:** periodic, lcg_simple, and lfsr_5 all achieve 100% test accuracy
2. **State space 2^{31} LCG is completely unlearnable:** lcg_glibc test accuracy equals random guessing (0.41% vs 0.39% baseline), while training accuracy is 100%—pure memorization, zero generalization
3. **LFSR-31 exhibits partial Grokking:** 50.16% accuracy is not random; the model learned some 1-bit pattern (see Section 3.2)
4. **CSPRNG is completely unlearnable:** Even training accuracy is only 1%—it cannot even memorize

3.2 Unexpected Finding: “Partial Grokking” in LFSR-31

The 50% accuracy of LFSR-31 is quite interesting.

The random baseline is 0.39% (256-class classification), not 50%. 50% means the model learned some binary pattern.

Possible explanations: - The model learned the pattern of the output’s **least significant bit** (LSB)
- Or learned some **parity** determination

This suggests Grokking is not all-or-nothing, but rather “**learn as much as you can**”—the model captured the simplest 1-bit pattern in the sequence, but couldn’t learn the remaining 7 bits.

Dimension analysis supports this explanation:

Sequence	Average Intrinsic Dimension	Interpretation
lcg_glibc	10-14 dimensions	Medium dimension, but learned nothing
lfsr_31	2-4 dimensions	Extremely low—did capture some simple structure
csprng	10-14 dimensions	Medium dimension, learned nothing

LFSR-31’s representation space collapsed to 2-4 dimensions, exactly matching the hypothesis of “learned 1-2 bit pattern.”

3.3 Training Dynamics Comparison

Sequence	Final Train Acc	Final Test Acc	Diagnosis
periodic	100%	100%	Perfect generalization
lcg_simple	100%	100%	Perfect generalization
lfsr_5	100%	100%	Perfect generalization
lcg_glibc	100%	0.41%	Pure memorization, zero generalization
lfsr_31	99.9%	50.2%	Partial generalization (1 bit)
csprng	1.0%	0.33%	Even memorization failed

lcg_glibc’s “**100% train + 0% test**” is a textbook “**memorization solution**”—the model memorized 3000 training sample input-output mappings, but there’s no generalizable structure between these mappings.

4. Theoretical Explanation

4.1 Reinterpreting with the Epiplexity Framework

The core formula from the Epiplexity paper:

$$P^* = \arg \min_{P \in \mathcal{P}_T} \{|P| + \mathbb{E}[\log 1/P(X)]\}$$

$$S_T(X) = |P^*| \quad (\text{Epiplexity})$$

In plain language: Within your computational budget, find the optimal balance between “shortest program + most accurate prediction.” The length of that program is the Epiplexity.

Explaining experimental results with this framework:

Sequence	State Space	Shortest Program Model Can Find	Epiplexity
periodic	7	$(i \% 7) > 3$	Very low (few bits)

Sequence	State Space	Shortest Program Model Can Find	Epiplexity
lcg_simple	256	$x = (3x + 7) \% 256$	Low (tens of bits)
lfsr_5	32	$x = x[-1] \text{ XOR } x[-5]$	Low
lcg_glibc	2^{31}	Cannot find	0
lfsr_31	2^{31}	$\text{LSB}(x) = \dots$	Very low (~ 1 bit)
csprng	∞	Cannot find	0

Epiplexity = 0 means: Within the model’s computational budget, there is no extractable structure in the data.

4.2 Physical Meaning of the Phase Transition Point

Experiments found the phase transition point between **256** and **2^{31}** .

This is not coincidental: - Model’s hidden dimension = 128 - 128-dimensional space can “comfortably” encode $2^7 = 128$ to $2^8 = 256$ states - But cannot encode 2^{31} states—off by 23 orders of magnitude

In other words: The model’s “memory capacity” is approximately between 2^7 and 2^8 ; beyond this, it can only generalize by “discovering patterns.”

- lcg_simple: 256 states, right at the boundary, can learn
- lcg_glibc: 2^{31} states, far beyond the boundary, cannot learn

4.3 Partial Grokking in LFSR-31: Learn What You Can

The LFSR-31 result suggests a more nuanced picture:

Grokking is not all-or-nothing, but layered.

The model may attempt to learn in order of increasing complexity: 1. First learn the simplest pattern (e.g., periodicity of LSB) 2. If successful, try to learn more complex patterns 3. If failed (insufficient computational power), stop at the current level

LFSR-31’s 50% accuracy = learned the first layer (1 bit), stuck at the second layer (remaining 7 bits).

This is consistent with the **two-stage Grokking** we discovered in our modular multiplication experiments: - Modular multiplication learned the quotient group $\mathbb{Z}_{12}$, not the complete $\mathbb{Z}_{96}$ - LFSR-31 learned the 1-bit pattern, not the complete 8-bit pattern

Unified explanation: Grokking is layered manifold discovery; each layer requires a different computational budget.

5. Connection to Grokking Experiments

This experiment complements our previous Grokking manifold discovery experiments (modular addition/multiplication):

Dimension	Grokking Experiments	This Experiment
Core question	What happens during Grokking?	What tasks can Grok?
Task type	Deterministic mathematical operations	Pseudo-random sequence prediction
Focus	Geometric changes in representation space	Boundary of learnability
Discovery	Dimension collapse, topological phase transition	State space determines learnability

Unified picture from both experiments:

1. **Grokking = Manifold Discovery:** From high-dimensional memorization curve to low-dimensional structural manifold
2. **Learnability = Whether Manifold Exists:** If there’s no low-dimensional manifold behind the data (like CSPRNG), Grokking cannot occur
3. **Partial Grokking = Partial Manifold Discovery:** Both LFSR-31 and modular multiplication only discovered “shallow” manifolds

6. Discussion

6.1 Implications for AI Training

1. **Value of Synthetic Data:** The Epiplexity framework proves that using AI-generated data to train AI is meaningful—computation can “unfold” structure folded in original rules
2. **Data Quality Assessment:** Epiplexity can quantify the “nutritional value” of datasets—high Epiplexity data can teach models more
3. **Model-Data Matching:** Not all data suits all models. Small models cannot learn patterns with large state spaces; forcing it only yields memorization solutions

6.2 Model Scale Experiment: Can Larger Models Break the Boundary?

To verify whether the learnability boundary is limited by model capacity, we conducted model scale ablation experiments:

Model	Parameters	embed_dim	layers	heads
Small	0.3M	128	2	4
4x	8M	512	4	8
10x	33M	1024	4	16

Results:

Sequence	Small (0.3M)	4x (8M)	10x (33M)
lcg_glibc	train 100%, test 0.4%	train 0.8%, test 0.5%	train 0.8%, test 0.5%
lfsr_31	train 100%, test 50%	train 100%, test 50%	train 0.8%, test 0.5%

Key Findings:

1. **lcg_glibc**: Larger models did not improve generalization; instead, weight decay caused underfitting (couldn't even memorize training set)
2. **lfsr_31**: 4x model maintained 50% partial generalization; but 10x model severely underfitted, couldn't even learn the 1-bit pattern
3. **Small model is optimal**: For this type of task, model too large + regularization = underfitting

Conclusion: The learnability boundary is a property of the task itself, not a model capacity issue. Scaling up models cannot break the boundary and may actually make things worse.

“It’s not that the model is too small; it’s that the pattern is too complex.”

6.3 Other Limitations

1. **No fine-grained exploration**: 256 and 2^{31} differ by 23 orders of magnitude; where exactly is the phase transition point?
2. **What is LFSR-31’s 1 bit**: We hypothesize it’s the LSB, but haven’t verified

6.4 Future Directions

1. **Fine-grained phase transition curve**: Test LCG with state spaces 2^{10} , 2^{15} , 2^{20}
2. **Remove weight decay**: Verify whether large models can break the boundary without regularization
3. **Dissecting LFSR-31**: Use Mechanistic Interpretability to find which bit the model learned

7. Conclusion

This paper experimentally validates the core prediction of the Epiplexity paper: **Information is observer-dependent**.

The same pseudo-random sequence: - For a cryptographer (who knows the algorithm): Epiplexity $\approx$ seed length - For a 2-layer Transformer: Epiplexity may be 0 (completely unlearnable)

We found the boundary of neural network learnability: - **State space \$ 256**: Learnable (100% test accuracy) - **State space = 2^{31}** : Unlearnable (test accuracy = random) - **LFSR-31**: Partially learnable (50% = learned 1 bit)

One sentence summary: Computational power determines what you can see in the data.

References

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“Shannon asked ‘how much information does this data objectively contain’; Epiplexity asks ‘given your capabilities, what can you learn from it.’”